

MSE 311 Thermodynamics of materials-Homework 8

Name:

Student ID:

1. The vapor pressure of solid NaF varies with temperature as

$$\ln p(\text{atm}) = \frac{-34,450}{T} - 2.01 \ln T + 33.74$$

and the vapor pressure of liquid NaF varies with temperature as

$$\ln p(\text{atm}) = \frac{-31,090}{T} - 2.52 \ln T + 34.66$$

Calculate

- a) The normal boiling temperature of NaF;
- b) The temperature and pressure at the triple point;
- c) The molar enthalpy of evaporation of NaF at its normal boiling temperature;
- d) The molar enthalpy of melting of NaF at the triple point;
- e) The difference between the constant-pressure molar heat capacities of liquid and solid NaF.

2. You are responsible for the purchase of oxygen gas which, before use, will be stored at a pressure of 200 atm at 300 K in a cylindrical vessel of diameter 0.2 m and height 2 m. Would you prefer that the gas behaved ideally or as a van der Waals fluid? The van der Waals constants for oxygen are $a = 1.36 \text{ liters}^2 \cdot \text{atm} \cdot \text{mole}^{-2}$ and $b = 0.0318 \text{ liters/mole}$. (Consider weight)

4. Derive the reduced equation of state for the Peng-Robinson's equation.